

Some recorded trees, such as the astonishing Gingko, are said to have lived thousands of years. But will we ever truly know if trees can be immortal? When a team of researchers from China and the US studied the growth rings from a sample of Gingko biloba trees, they discovered something amazing. The older trees were just as healthy as the younger ones. Researchers studied a total of 34 Gingko biloba trees, ranging from three to 667 years old. They looked for age-related changes in the cambium, a layer of dividing cells beneath the trees' bark that generates growth of stems and roots after the first season. They discovered that leaf size, photosynthetic ability, and seed quality – which are all indicators of health – didn't worsen with age. In an earlier study of nearly 700,000 trees, scientists found that a tree's growth doesn't slow down after hundreds of years. In fact, it speeds up. Over time, trees stopped getting taller but, like bodybuilders, they did get wider. So, does all this mean that trees can live forever? Well, that's not an easy question to answer. Whilst a 667-year life span is incomprehensible to humans – and even more so for the delicate mayfly, which lives for only a day – it's relatively young for a Gingko. Amazingly, they can live for thousands of years. But while signs of deterioration from old age in trees might not be perceptible to humans in our lifetime, this doesn't necessarily mean they're immortal. Stress and strain are likely to eventually take their toll, but studies would have to take place over hundreds – or thousands – of years to say for certain. One possible sign of age-related deterioration was seen by the researchers studying the Gingkoes. They found that the older trees had thinner vascular tissue... but also concluded that the vascular cambium was capable of growth for hundreds of years, or millennia. And so, saying for definite whether trees do or do not die from old age requires more data, and the problem with that is long-lived trees are pretty hard to come by.